# Assignments-1

**Aim**: Design and implement a simple reflex agent for the Vacuum Cleaner world and analyze its rational behavior in a given environment.

## Learning Objectives

After completing this practical, students will be able to:

1. Understand the concept of an **intelligent agent**.
2. Implement a **Simple Reflex Agent**.
3. Understand **Percept, Action, and Agent Function**.
4. Analyze the rational behavior of an agent in a given environment.

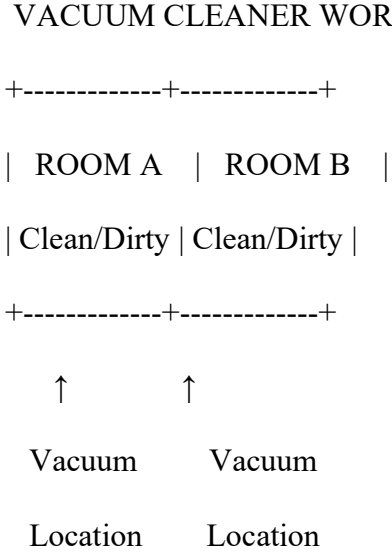
## Problem Statement

Consider a two-room Vacuum Cleaner World consisting of **Room A** and **Room B**. Each room can be either **Clean** or **Dirty**. The vacuum cleaner can be located in either room.

The agent follows these rules:

| Current Location | Room Status | Action |
|---|---|---|
| A | Dirty | Suck |
| A | Clean | Move Right |
| B | Dirty | Suck |
| B | Clean | Move Left |

The agent should select an appropriate action based only on the **current percept**.



## Simple Reflex Agent Algorithm

**Input:** Current location and status of the current room.

**Algorithm:**

1. Read the current percept.
2. If the current room is dirty, perform **SUCK**.
3. If the current room is clean:
    - If the vacuum is in Room A, move to Room B.
    - If the vacuum is in Room B, move to Room A.
4. Repeat until the required condition is satisfied.

**Python Program:**

```
# Simple Reflex Agent for Vacuum Cleaner World


def vacuum_agent(location, status):

    if status == "Dirty":

        return "Suck"


    elif location == "A":

        return "Move Right"


    elif location == "B":

        return "Move Left"




# Given environment

location = input("Enter vacuum location (A/B): ")
```

```
status = input("Enter room status (Clean/Dirty): ")


action = vacuum_agent(location, status)


print("Current Location:", location)

print("Room Status:", status)

print("Agent Action:", action)
```

## Analysis of Rational Behavior:

A **rational agent** selects an action that maximizes its expected performance based on the percept available to it.

For the Vacuum Cleaner World:

- If the current room is **Dirty**, the rational action is **Suck** because cleaning increases the environment's cleanliness.
- If the current room is **Clean**, the agent moves to the other room to check whether it is dirty.
- The agent does not need to know the complete state of the environment.
- The agent uses a **condition-action rule** to select its action.

## PEAS Description

| Component | Description |
|---|---|
| **P – Performance Measure** | Maximum cleanliness, minimum unnecessary movement |
| **E – Environment** | Two rooms: A and B |
| **A – Actuators** | Suck, Move Left, Move Right |
| **S – Sensors** | Detect current location and room cleanliness |

## Conclusion:

Thus, a Simple Reflex Agent for the Vacuum Cleaner World was successfully designed and implemented. The agent selects rational actions based on the current percept using condition-action rules.

**Out put:**

In [1]:
```
def vacuum_agent(location, status):
    if status == "Dirty":
        return "Suck"

    elif location == "A":
        return "Move Right"

    elif location == "B":
        return "Move Left"



# Given environment
location = input("Enter vacuum location (A/B): ")
status = input("Enter room status (Clean/Dirty): ")

action = vacuum_agent(location, status)

print("Current Location:", location)
print("Room Status:", status)
print("Agent Action:", action)
```

```
Enter vacuum location (A/B): B
Enter room status (Clean/Dirty): Dirty
Current Location: B
Room Status: Dirty
Agent Action: Suck
```